

Local Search for Job Shop under Interval Uncertainty

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Abstract

1. Introduction

1.1. Uncertain Processing Times and Due Dates

(Camino:) Esto es literal del paper postHAIS As argued in Section ??, in real life applications is common to encounter uncertainty regarding the exact time it will take to process a task on the final due date for a job. If only an upper and a lower bound of each duration and due date are known, uncertainty can be represented as a closed interval $\mathbf{a} = [a^-, a^+] = \{x \in \mathbb{R} : a^- \leq x \leq a^+\}$, with a^- and a^+ the corresponding lower and upper bounds.

Let $\mathbb{IR}$ denote the set of closed intervals. The job shop problem with total tardiness minimisation requires three arithmetic operations on $\mathbb{IR}$: addition, subtraction and maximum. These are defined by extending the corresponding operations on real numbers [1], so given two intervals $\mathbf{a} = [a^-, a^+]$, $\mathbf{b} = [b^-, b^+] \in \mathbb{IR}$, the addition is expressed as $[a^- + b^-, a^+ + b^+]$, the subtraction as $[a^- - b^+, a^+ - b^-]$ and the maximum as $[\max(a^-, b^-), \max(a^+, b^+)]$.

Comparisons are a key point in scheduling, as the “best” schedule should be the one with “minimal” total tardiness (an interval). However, there is no natural total order in the set of intervals, so an interval ranking method needs to be considered among those proposed in the literature [2]. Among the multiple

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existing rankings in $\mathbb{IR}$, here we consider the following:

$$\mathbf{a} \leq_{Lex1} \mathbf{b} \Leftrightarrow a^- < b^- \vee (a^- = b^- \wedge a^+ \leq b^+), \quad (1)$$

$$\mathbf{a} \leq_{Lex2} \mathbf{b} \Leftrightarrow a^+ < b^+ \vee (a^+ = b^+ \wedge a^- \leq b^-), \quad (2)$$

$$\mathbf{a} \leq_{YX} \mathbf{b} \Leftrightarrow a^- + a^+ < b^- + b^+ \vee (a^- + a^+ = b^- + b^+ \wedge a^+ - a^- \leq b^+ - b^-), \quad (3)$$

$$\mathbf{a} \leq_{MP} \mathbf{b} \Leftrightarrow m(\mathbf{a}) \leq m(\mathbf{b}) \text{ with } m(\mathbf{a}) = (a^- + a^+)/2. \quad (4)$$

It can be seen that (1), (2) and (3) actually define total order relations in $\mathbb{IR}$ [3]. Both (1) and (2) are derived from a lexicographical order of interval extreme points. The ranking in expression (3) is proposed in [4], and the last one (mid-point order) is a particular case of the classical Hurwitz criterion and is equivalent to the one used in [5] for interval JSP.

(Camino:) Esto es nuevo. Se necesita para las demostraciones posteriores. If $a^- \leq b^-$ and $a^+ \leq b^+$ then $\mathbf{a} \leq_R \mathbf{b}$ for all previously defined rankings.

Pongo las demostraciones, aunque si son triviales se quitan (Revision: la de MP es trivial y el resto supongo que no se necesitan)

(Camino:) EN lo siguiente no he cambiado a nuestra notacion de intervalos porque estoy reproduciendo literalmente del paper de Bustince para que decidamos que poner y como From [3]: For each interval $[a, b] \subset \mathbb{R}$ we can assign a niquely point $(a, b) \in \mathbb{R}^2$ and that intervals can be ordered by means of orders of points in $\mathbb{R}^2$. The usual partial order in $\mathbb{R}^2$, given by $(a, b) \leq_2 (c, d) \iff a \leq c \wedge b \leq d$ induces the partial order of intervals which will be denoted by $\leq_2$,

$$[a, b] \leq_2 [c, d] \iff a \leq c \wedge b \leq d$$

. For the set of closed intervals in $[0, 1]$, $L([0, 1])$, given a poset $(L([0, 1]), \leq)$, the relation between intervals $\preceq$ is an order admissible is is a linear order on $L([0, 1])$ and refines the order $\leq_2$. That means that if $a^- \leq b^-$ and $a^+ \leq b^+$ then $\mathbf{a} \preceq \mathbf{b}$ for all intervals in $[0, 1]$. The rankings $\leq_{Lex1}$, $\leq_{Lex2}$, $\leq_{YX}$ and, obviously, the order $\leq_2$ ($[a, b] \leq_2 [c, d]$ iff $a \leq c$ and $b \leq d$).

(Camino:) Supongo que es natural generalizar el concepto de orden admisible para intervalos acotados que no esten en $[0, 1]$. De hecho, supongo que estar en algn paper, pero no lo encuentro. Si es as las propiedades que necesitamos para analizar la vecindad son ciertas para todos los ordenes admisibles, entre los que se cuentan 3 de los 4 rankings definidos, y tambien para el MP, que no es un orden, pero si cumple la propiedad que buscamos, es decir, que extiende el orden $\leq_2$, es decir que

$$[a, b] \preceq [c, d] \text{ whenever } [a, b] \leq_2 [c, d]$$

(Camino:) Lo siguiente, por tanto, sobraria

Para Lex1: Suponemos $a^- \leq b^-$ and $a^+ \leq b^+$, como $a^- \leq b^-$,

- si $a^- < b^-$ entonces $\mathbf{a} \leq_{Lex1} \mathbf{b}$
- si $a^- = b^-$, como $a^+ \leq b^+$ de nuevo $\mathbf{a} \leq_{Lex1} \mathbf{b}$

Para Lex2 la demostracion es analoga

Para YX: Suponemos $a^- \leq b^-$ and $a^+ \leq b^+$, $a^- + a^+ \leq b^- + b^+$,

- si $a^- + a^+ < b^- + b^+$ entonces $\mathbf{a} \leq_{YX} \mathbf{b}$
- si $a^- + a^+ = b^- + b^+$, suponemos (RA) que $a^+ - a^- > b^+ - b^-$. Usando la igualdad y sustituyendo a^- en la desigualdad $a^+ - (b^+ + b^- a^+) > b^+ - b^-$ y, en consecuencia, $2a^+ > 2b^+$ que es imposible con la hiptesis $a^+ \leq b^+$; de nuevo $\mathbf{a} \leq_{YX} \mathbf{b}$

Para MP es trivial: si $a^- \leq b^-$ and $a^+ \leq b^+$, $(a^- + a^+) \leq (b^- + b^+)$ luego, $\mathbf{a} \leq_{MP} \mathbf{b}$.

2. Criticality

En deterministic scheduling ES_i = longitud del camino más largo de 0 a i (*earliest starting time of i*) LS_i = longitud del camino más largo longitud de camino más largo de $i+1$ al node *end* (*latest starting time*) $F_i = LS_i - ES_i$ (*float of i*)

Standard longest path computations in acyclic graph allow to compute all these values in $O(|E|)$. All tasks of critical path have null float.

When problem parameters are ill-known, a common way to modelling uncertainty is to define each task processing time as an interval $\mathbf{p}_i = [p_i^-, p_i^+] = \{x \in \mathbb{R} : p_i^- \leq x \leq p_i^+\}$, with p_i^- and p_i^+ the corresponding minimum and maximum values for this processing time.

Temporal analysis focuses on the computation of the minimum and maximum values of the earliest start times, latest start time and total floats.

Definition 1. A configuration ω is a tuple $\omega = (p_1, \dots, p_o)$ of durations such that $\forall i \in O, p_i \in \mathbf{p}_i$.

Ω denotes the set of all configurations. The processing time of task i in configuration ω is denoted $p_i(\omega)$. A configuration defines an instance of the corresponding deterministic scheduling problem.

Using configurations, the possible values of ES_i , LS_i for the earliest and latest starting times and F_i for float are the closed intervals defined by the following bounds:

- Best-case (minimum, lower bound) earliest starting time $ES_i^- = \min_{\omega \in \Omega} ES_i(\omega)$.
- Worst-case (maximum, upper bound) earliest starting time $ES_i^+ = \max_{\omega \in \Omega} ES_i(\omega)$.
- Worst-case (minimum, lower bound) latest starting time $LS_i^- = \min_{\omega \in \Omega} LS_i(\omega)$.
- Best-case (maximum, upper bound) latest starting time $LS_i^+ = \max_{\omega \in \Omega} LS_i(\omega)$.
- Worst-case (minimum, lower bound) float $F_i^- = \min_{\omega \in \Omega} F_i(\omega)$.
- Best-case (maximum, upper bound) float $F_i^+ = \max_{\omega \in \Omega} F_i(\omega)$.

These values are of interest for providing valuable information to the decision-maker about the level of criticality of the tasks despite the uncertain nature of processing times.

Following [6], a task i is *possibly critical* if there exists a configuration $\omega \in \Omega$ in which i is critical in the usual sense, which is equivalent to that the lower bound of its float is null, that is, $F_i^- = 0$. Special attention should be paid for *possibly critical* tasks for risk-adverse decision policies.

A task i is *necessarily critical* if is critical in the usual sense in all configurations $\omega \in \Omega$, which is equivalent ([6]) to that the upper bound of its float is null, that is, $F_i^+ = 0$. A *necessarily critical* task should be carefully monitored independently of the scenario.

Is a task is *necessarily critical* then $ES_i = LS_i$. However, if $ES_i = LS_i$, the task is only *possibly critical*.

Several algorithms to compute critical tasks use the notion of *extreme scenario* induce by a task subset. Let $\omega^+(Q)$, with $Q \subseteq V$, the extreme scenario such that each task $i \in Q$ is set to p_i^+ while each activity of $i \in V \setminus Q$ is set to p_i^- . Notice that the earliest starting times of each task $i \in V$ are attained on extreme scenarios induced by the empty set and V respectively:

$$ES_i^- = ES_i(\omega^+(\emptyset))$$

and

$$ES_i^+ = ES_i(\omega^+(V)).$$

Furthermore, for each task $i \in V$ it holds that the latest starting times and floats are attained for extreme scenarios [7] (yo lo he visto en [8]):

$$LS_i^+ = \max_{Q \subseteq V} LS_i(\omega^+(Q))$$

$$LS_i^- = \min_{Q \subseteq V} LS_i(\omega^+(Q))$$

$$F_i^+ = \max_{Q \subseteq V} F_i(\omega^+(Q))$$

$$F_i^- = \min_{Q \subseteq V} F_i(\omega^+(Q))$$

In fact, subsets Q can be limited to subsets of tasks located on paths from node *start* to node *end* [6].

To compute possibly and necessarily critical paths, a purely enumerative algorithm would enumerate all 2^o extreme scenarios and, for each of them, compute the required paths. In the particular case of minimum float, it remains a NP-hard problem [8] and the most efficient algorithm to compute it is a branch-and-bound. This is not useful in the framework of local search algorithms where the critical paths have to be computed a huge number of times.

In order to simplify the critical path computations and to define a neighbourhood structure that holds some of the highly desirable properties of neighbourhoods, we will adapt the idea proposed in [9] for Fuzzy Job Shop to the interval case. Thus, given a *feasible processing order* $\sigma \in \Sigma$ and its corresponding solution graph $G(\sigma) = (V, A \cup D(\sigma))$, we denote $G^-(\sigma)$ and $G^+(\sigma)$ the solution

graphs which are identical to $G(\sigma)$ except for the cost of each arc (x, y) that is p_x^- and p_x^+ respectively. These are the directed acyclic graphs defined by $D(\sigma)$ in the disjunctive graphs representing the extreme scenarios induced by the empty set and V respectively. In coherency, we will named extreme solution graphs.

Definition 2. *A path P in $G(\sigma)$ is extreme critical if and only if P is critical in $G^-(\sigma)$ or in $G^+(\sigma)$. Nodes, arcs and blocks in a extreme critical path are critical.*

Notice that, all tasks on critical paths in $G^-(\sigma)$ and $G^+(\sigma)$ are possibly critical and contain the set of necessarily critical tasks.

Proof If i is critical in $G^-(\sigma)$ or in $G^+(\sigma)$, then $F_i(\omega^+(\emptyset)) = 0$ or $F_i(\omega^+(V)) = 0$. Thus, $F_i^- = \min_{Q \subseteq V} F_i(\omega^+(Q)) \leq \min\{F_i(\omega^+(\emptyset)), F_i(\omega^+(V))\} = 0$, so i is possibly critical.

And, if i is necessarily critical, $F_i^+ = 0$, then $F_i^+ = \max_{Q \subseteq V} F_i(\omega^+(Q)) \geq \max\{F_i(\omega^+(\emptyset)), F_i(\omega^+(V))\}$ and then $F_i(\omega^+(\emptyset)) = 0 = F_i(\omega^+(V))$, and therefore, i is critical both in $G^-(\sigma)$ and $G^+(\sigma)$. **endproof**

Given a feasible processing order σ and the schedule $\mathbf{s}(\sigma)$ built over σ (supongo que podemos poner, en otro sitio, la definicion de una planificacion que respete el orden de procesamiento y que asigne como tiempo de inicio de cada tarea el maximo entre el tiempo de fin de su predecesora en el trabajo y la predecesora en la maquina definida por σ), the makespan $[C_{max}^-, C_{max}^+]$ of the schedule is not necessarily the cost of a extreme critical path in $G(\sigma)$, but C_{max}^- and C_{max}^+ are the cost of a critical path in $G^-(\sigma)$ and in $G^+(\sigma)$ respectively.

3. Neighbourhood

We start extending to the interval job shop, the neighbourhood structure defined in [10] for the JSP and extended for fuzzy JSP in [9].

Definition 3. *Let $G(\sigma) = (V, A \cup D(\sigma))$ be a solution graph and let $\sigma_{(v)}$, with $v \in D(\sigma)$, be the processing order built from σ after reverse the processing order of tasks in arc v . The neighbourhood of the solution represented by σ is defined by*

$$H(\sigma) = \{\sigma_{(v)} : v \in D(\sigma) \text{ is extreme critical}\} \quad (5)$$

If we denote with

$$H'(\sigma) = \{\sigma_{(v)} : v \in D(\sigma) \text{ is possibly critical}\}$$

is it obvious that $\forall \sigma \in \Sigma, H(\sigma) \subset H'(\sigma)$ and it is a proper subset (PONER UN EJEMPLO). This reduction in the size of the set of neighbours, which is an interesting property of the neighbourhood, also allows to compute this set more efficiently, which makes it available for local search algorithms and we will see that does not affect the potential quality of solutions found by the local search whatever the ranking used to compare intervals, between the proposed rankings, that is, never improving neighbours can be lost.

Proposition 1. *Let $\sigma \in \Sigma$ be a feasible processing order and let $\sigma_{(v)} \in \Sigma$ be a feasible processing order obtained by the reversal an arc v which is not extreme critical in $G(\sigma)$. Then*

$$C_{max}^-(\sigma) \leq C_{max}^-(\sigma_{(v)}), \text{ and } C_{max}^+(\sigma) \leq C_{max}^+(\sigma_{(v)}) \quad (6)$$

and therefore, for every ranking R (de los considerados aqui)

$$\mathbf{C}_{\max}(\sigma) \leq_R \mathbf{C}_{\max}(\sigma_{(v)}) \quad (7)$$

Proof Let $\sigma_{(v)}$ be obtained from σ by the reversal of an arc $v = (x, y)$ where v is not a extreme critical arc, that is, it is not a critical arc in $G^-(\sigma)$ nor $G^+(\sigma)$. Clearly, the critical arcs of $G^-(\sigma)$ remain unchanged in $G^-(\sigma_{(v)})$ after the move, and therefore $C_{max}^-(\sigma) \leq C_{max}^-(\sigma_{(v)})$ and analogous reasoning is valid for $G^+(\sigma)$. $\square$

In particular, neighbours in H' which do not belong to H can never improve the makespan regardless the ranking used to compare intervals (otra vez esto solo para los rankings considerados):

Corollary 1. $\forall \sigma \in \Sigma, \forall \tau \in (H'(\sigma) - H(\sigma)) \cap \Sigma, \mathbf{C}_{\max}(\sigma) \leq_R \mathbf{C}_{\max}(\tau)$

(Camino:) Creo que las demostraciones de facibilidad y conectividad de la estructura H se va a poder demostrar como en fuzzy. Cruzo los dedos y VOY PROBANDO. Hay que reescribirlo, porque adapto la notacion, pero no superaria el urkund

Just as it happened in fuzzy JSP and unlike the crisp csse, the makespan of the schedule may not be the cost of a critical path. However, in each bound of the interval that represents the makespan, $C_{max}^*(\sigma), * \in \{-, +\}$ is the cost of a critical path in the corresponding solution extreme graph $G^*(\sigma)$.

For a solution graph $G(\sigma)$ and a task x , let $\mathbf{s}_x$ and $\mathbf{c}_x$ denote respectively the starting and completion times of x , let $P\nu_x$ and $S\nu_x$ denote the predecessor and successor nodes of x on the machine sequence (predecessor and successor in $D(\sigma)$), and let PJ_x and SJ_x denote respectively the predecessor and successor nodes of x on the job sequence (predecessor and successor in A). We now extend some well-known notions for the crisp job shop and define the *head* $\mathbf{r}_x$ of task x as the starting time of x i.e. $\mathbf{s}_x$, in our context, a TFN $\mathbf{r}_x = [r_x^-, r_x^+]$ given by:

$$\mathbf{r}_x = \max\{\mathbf{r}_{PJ_x} + \mathbf{p}_{PJ_x}, \mathbf{r}_{P\nu_x} + \mathbf{p}_{P\nu_x}\}, \quad (8)$$

and the *tail* $\mathbf{q}_x$ of task x as:

$$\mathbf{q}_x = \max\{\mathbf{q}_{SJ_x} + \mathbf{p}_{SJ_x}, \mathbf{q}_{S\nu_x} + \mathbf{p}_{S\nu_x}\}, \quad (9)$$

Clearly, $\mathbf{c}_x = \mathbf{r}_x + \mathbf{p}_x$. For each extreme graph $G^*(\sigma), * \in \{-, +\}$, r_x^* is the length of the longest path from node *start* to node x and $q_x^* + p_x^*$ is the length of the longest path from node x to node *end*. Hence, $r_x^* + p_x^* + q_x^*$ is the length of the longest path from node *start* to node *end* through node x in $G^*(\sigma)$; it is a lower bound of the corresponding extreme of the interval of the makespan,

being equal to $C_{max}^*(\sigma)$ if node x belongs to a critical path in $G^*(\sigma)$. For two intervals $\mathbf{a}$ and $\mathbf{b}$ let $\mathbf{a} \simeq \mathbf{b}$ denote that one of the bounds of the intervals are equal: $a^- = b^- \vee a^+ = b^+$. The next proposition states some properties of critical arcs and tasks which follow trivially from the above definition:

Proposition 2. *If an arc (x, y) is critical, then $\mathbf{r}_x + \mathbf{p}_x \simeq \mathbf{r}_y$. A task x is critical if and only if $\mathbf{r}_x + \mathbf{p}_x + \mathbf{q}_x \simeq \mathbf{C}_{max}$.*

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